Skill Forecasting and Semantic Career Pathway Optimization Ecosystem for Undergraduates

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Project Proposal Report

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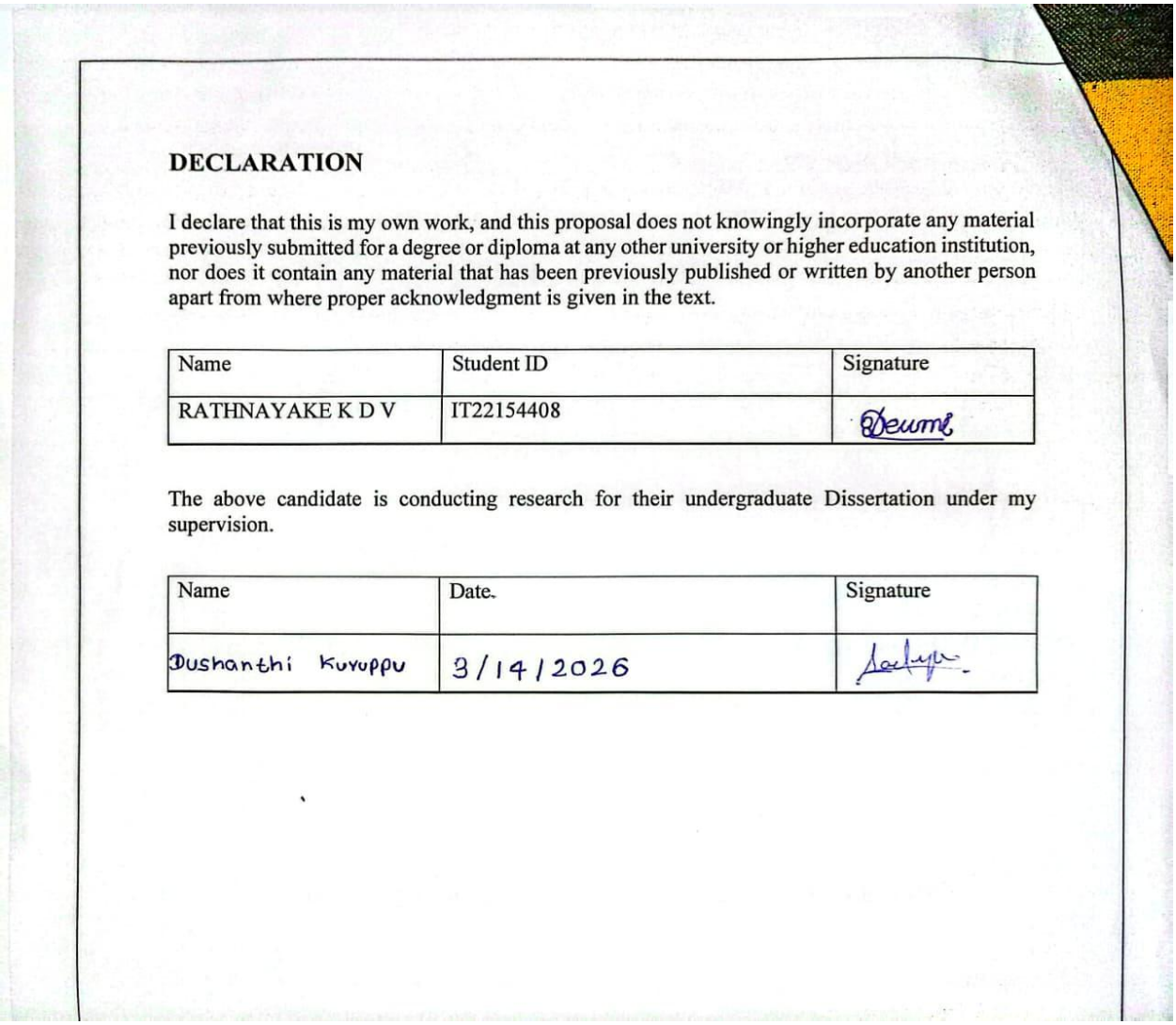
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Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka.

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DECLARATION



DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The rapid growth of the Information Technology (IT) sector has created a gap between university programs and industry skill needs in Sri Lanka. IT students face a lot of uncertainty in figuring out which technical skills will be important by the time they graduate. This uncertainty leads to underemployment and longer corporate retraining periods. A survey of 320 students showed that 89.4% feel a mismatch between their curriculum and industry needs. Additionally, 87.5% find it hard to predict future skill demands, and all students would modify their learning plans if they had reliable forecasts. Current career platforms only offer retrospective snapshots and do not analyze image-based job postings or use global trends to make predictions.

This research aims to fill these gaps by creating a Semantic Skill Forecasting Engine, which serves as the predictive foundation of an AI-powered career ecosystem. The engine uses OpenCV and Tesseract OCR to convert unstructured job postings into digital format. It applies BERTopic for semantic skill grouping and ARIMA time-series models to produce three-month demand forecasts. A unique Global-Local Lead-Lag Analysis uses cross-correlation and Granger causality to measure time delays between global and Sri Lankan skill adoption, providing early warnings of new trends.

The system is built in Python, using statsmodels, Hugging Face Transformers, and FastAPI. It has been validated with a Character Error Rate of less than 10%, clustering precision and recall of 85% or more, forecast accuracy with a MAPE of less than 20%, and a System Usability Scale score of 70 or above. The anticipated outcomes include a working forecasting engine, an interactive dashboard, and a validated method for analyzing skills across different markets. The platform supports SDG 4 (Quality Education) and SDG 8 (Decent Work) and has commercial potential through subscriptions from students and licensing from universities.

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TABLE OF CONTENT

Contents

DECLARATION 2

ABSTRACT 3

ACKNOWLEDGEMENT 4

TABLE OF CONTENT 5

LIST OF TABLES 7

LIST OF FIGURES 8

LIST OF ABBREVIATIONS 9

LIST OF APPENDICES 10

INTRODUCTION 11

1.1 Background and Context 11

1.1.1 The Real-World Problem and Affected Parties 11

1.1.2 Statistical Evidence and Primary Validation 11

1.1.3 Analysis of Existing Solutions and Their Limitations 12

1.2 Literature Review 13

1.2.1 Existing Research Approaches 14

1.2.2 The Spatial Dimension: Global vs. Local Markets 15

1.2.3 Review of Commercial Products and Services 15

1.2.4 Synthesis of Identified Limitations and Research Gap 15

1.2.5 Comparison of Existing Literature and Proposed System 16

1.2.6 Conclusion of Literature Review 17

1.3 Research Gap 17

1.3.1. The Semantic Extraction Gap 17

1.3.2. The Temporal Forecasting Gap 18

1.3.3. The Spatial Analysis Gap (Global vs. Local) 18

1.3.4 Justification of Novelty and Creativity 18

2. OBJECTIVE 19

2.1 Main Objective 19

2.2 Specific Objectives 19

3.METHODOLOGY 21

3.1 Research Approach 21

3.3 Tools and Platforms with Justification 22
3.4 Data Collection and Ethical Considerations 22
3.5 Validation Strategy and Performance Metrics 23
4. HIGH-LEVEL SYSTEM ARCHITECTURE 24
4.1 Overview of Proposed Solution 24
4.2 System Diagram 25
4.3 Highlight of Individual Component and Its Parts 25
4.4 Component Interactions 26
4.5 Data Flow Explanation 26
4.6 Scalability and Security Considerations 27
5. USER REQUIREMENTS 28
5.1 Stakeholder Identification 28
5.2 Functional Requirements 28
5.3 Non-functional Requirements 29
6. COMMERCIALIZATION PLAN 29
7. BUDGET AND JUSTIFICATION 31
8. WORK BREAKDOWN STRUCTURE (WBS) 33
9. GANTT CHART 34
10. REFERENCES LIST 35
References 35

LIST OF TABLES

TABLE 1 LIST OF ABBREVIATIONS 9
TABLE 2 LIST OF APPENDICES 10
TABLE 3:SYNTHESIS OF IDENTIFIED LIMITATIONS AND RESEARCH GAP 16
TABLE 4 COMPARISON OF EXISTING LITERATURE AND PROPOSED SYSTEM 16
TABLE 5 SMART BREAKDOWN OF SPECIFIC OBJECTIVES 20
TABLE 6 APPLICATION IN THIS RESEARCH 21
TABLE 7 TECHNICAL VALIDATION METRICS 21
TABLE 8 TOOLS AND PLATFORMS WITH JUSTIFICATION 22
TABLE 9 DATA COLLECTION 22
TABLE 10 ETHICAL CONSIDERATIONS 23
TABLE 11 TECHNICAL VALIDATION METRICS 23
TABLE 12 COMPONENT INTERACTIONS 26
TABLE 13 NON-FUNCTIONAL REQUIREMENTS 29
TABLE 14 REVENUE MODELS 30
TABLE 15 ESTIMATED COSTS 30
TABLE 16 BUDGET SUMMARY 32

LIST OF FIGURES

FIGURE 0-1 EVIDENCE OF CURRICULUM–INDUSTRY MISMATCH AMONG SRI LANKAN IT UNDERGRADUATES (N = 320) 11
FIGURE 0-2 JOB SEARCH ACTIVITY AMONG RESPONDENTS 12
FIGURE 0-3 SURVEY RESPONDENT DEMOGRAPHICS – YEAR OF STUDY 12
FIGURE 0-4 HIGH-LEVEL SYSTEM ARCHITECTURE 24
FIGURE 0-5 DATA-DRIVEN SKILL FORECASTING & PATHWAY SYSTEM 25
FIGURE 0-18. WORK BREAKDOWN STRUCTURE (WBS) DIAGRAM 33
FIGURE 0-1 GANTT CHART 34

LIST OF ABBREVIATIONS

Table 1 LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
ARIMA	AutoRegressive Integrated Moving Average
BERT	Bidirectional Encoder Representations from Transformers
BERTopic	BERT-based Topic Modeling
CCF	Cross-Correlation Function
CER	Character Error Rate
DSR	Design Science Research
ETS	Exponential Smoothing (Error, Trend, Seasonal)
ICTA	Information and Communication Technology Agency of Sri Lanka
IT	Information Technology
KSA	Knowledge, Skills, and Abilities
LLM	Large Language Model
LSTM	Long Short-Term Memory
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
NLP	Natural Language Processing
OCR	Optical Character Recognition
PII	Personally Identifiable Information
RMSE	Root Mean Square Error
SDG	Sustainable Development Goal
SUS	System Usability Scale
TF-IDF	Term Frequency – Inverse Document Frequency
UI	User Interface
WBS	Work Breakdown Structure

LIST OF APPENDICES

Table 2 LIST OF APPENDICES

Appendix	Description
A	Survey Instruments (questionnaire and response summaries)
B	Detailed System Diagrams (architecture, data flow, database schema)
C	Preliminary UI Mockups (dashboard, career graph, CV optimizer, interview simulator)
D	Risk Analysis (risk register and mitigation strategies)
E	Ethical Documents (data collection ethics statement, ethics review approval)

INTRODUCTION

1.1 Background and Context

1.1.1 The Real-World Problem and Affected Parties

Information Technology (IT) is an industry that is highly recognized for its rapid and constant technological development. However, in developing countries like Sri Lanka, it is identified that university curricula face a huge problem in coping with the rapid development of IT and technology trends identified in global industries. This, in return, results in a very critical **skill mismatch**. The group of students most affected by this skill mismatch is undergraduate IT students, particularly those in their third and final year of study, as they step from academia into the real world. Once IT graduates enter the real world, they realize that there is a skill mismatch between their learned skills and skills identified as necessary by employers in this new age. This results in very high rates of unemployment for IT graduates and a talent gap for IT companies in Sri Lanka that aim to compete globally.

1.1.2 Statistical Evidence and Primary Validation

This skill gap is a recognized national challenge. According to the ICTA National IT BPM Workforce Survey [1], Sri Lanka aims to significantly expand its IT workforce. However, recent 2025 studies indicate that emerging paradigms, such as Artificial Intelligence (AI) and Cloud Computing, are rapidly altering employer expectations, leaving many local graduates unprepared [2].

To validate the severity of this problem locally, our research group conducted a primary survey among 320 Sri Lankan IT undergraduates. The empirical evidence strongly confirms the crisis. According to Question of the survey, a staggering 89.4% of the students reported experiencing a mismatch between their university curriculum and the requirements listed in current job advertisements with 35.6% encountering this frequently or always.

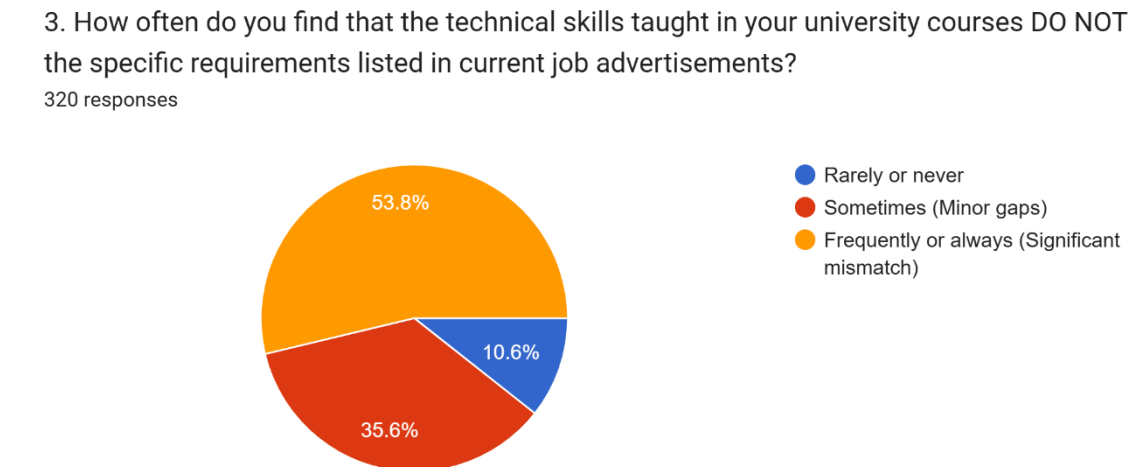


Figure 0-1 Evidence of curriculum–industry mismatch among Sri Lankan IT undergraduates (n = 320)

Furthermore, 75% of these respondents are in their 3rd or 4th year, and 56% are actively applying for jobs (Survey Questions 1 and 2). This statistical evidence proves that the problem is not theoretical. It is an immediate and critical hurdle for the majority of current IT students.

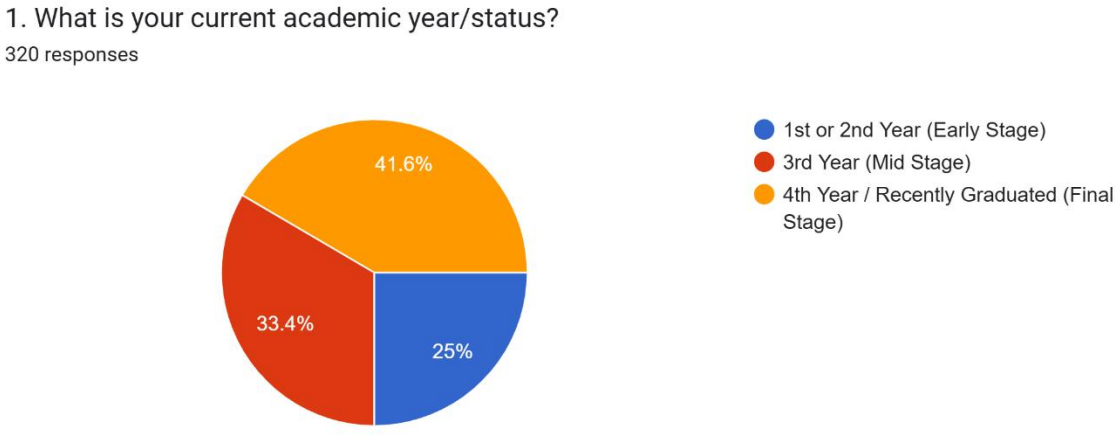


Figure 0-3 Survey respondent demographics – year of study

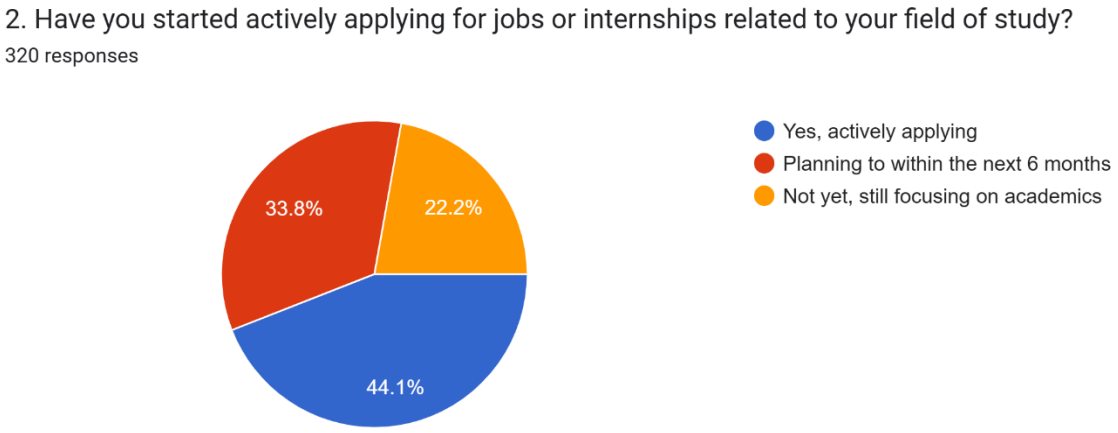


Figure 0-2 Job search activity among respondents

1.1.3 Analysis of Existing Solutions and Their Limitations

Currently, undergraduate students use general job portals such as TopJobs [3] and LinkedIn to manually identify the skills required. These portals only list the currently vacant positions and do not offer any upskilling strategies. In the academic domain, several local studies were conducted to analyze the trends of IT jobs by applying text mining techniques. Researchers applied techniques such as Term Frequency-Inverse Document Frequency (TF-IDF) and Apriori algorithms to extract skills from images of job vacancies [4] [5].

However, a significant knowledge gap exists. Firstly, existing approaches are based on rigid syntactic keyword matching. Such an approach is incapable of capturing the semantic context of unstructured job descriptions (for example, NodeJS and Express are not related to the same technology group). Secondly, existing research is heavily restricted by Static Descriptive Analytics. Existing research is only able to describe what skills have been demanded in the past. Moreover, existing research completely lacks short-term momentum forecasting . Lastly, existing local research is isolated. It does not take into consideration the fact that the IT industry is a highly globalized industry. In fact, the trends of remote jobs from abroad are predictive of future local demands.

Ideas for the Proposed Solution

To bridge this identified research gap, this project proposes. This is a comprehensive ecosystem for Skill Forecasting and Semantic Career Pathway Optimization. The necessity of a unified ecosystem. The ecosystem will have four main pillars: a predictive career graph, emotion-aware interview simulator, CV semantic matcher, and skill forecasting engine.

As the foundational data engine for this ecosystem, my individual component is on Semantic Skill Forecasting. To bridge the gap identified by using keyword matching, I propose using BERTopic for deep learning-based semantic skill extraction from text and image-based job postings. To bridge the gap identified by using static analytics, I will implement short-term market momentum forecasting using ARIMA and Exponential Smoothing algorithms. The scientific novelty of this component is the introduction of a global-local lead-lag analysis. By analyzing foreign remote job datasets and comparing them with local Sri Lankan datasets, the system will mathematically calculate lag times, enabling the AI to warn undergraduates on emerging global skills before they become standard requirements in the local Sri Lankan job market.

SDG Alignment

This proposed research closely aligns with the United Nations Sustainable Development Goals (SDGs).

- **SDG 4 (Quality Education):** It promotes lifelong learning by providing students with data-driven, real-time insights to self-correct their learning paths outside the traditional classroom.
- **SDG 8 (Decent Work and Economic Growth):** By directly addressing the root cause of graduate underemployment and optimizing candidate readiness, the ecosystem supports productive employment and strengthens the human capital of Sri Lanka's IT sector.

1.2 Literature Review

The rapid acceleration of digital evolution has resulted in a significant diminution of the skill half-life in the Information Technology (IT) industry, thereby increasing the requirement for skill

intelligence in anticipation. This section critically discusses the extant body of academic and commercial works on IT skill demand, including methodological approaches, applications, and limitations, which are directly related to the requirement of the proposed Semantic Forecasting Engine.

1.2.1 Existing Research Approaches

Recent scholarly efforts to analyze job market demands can be categorized into semantic skill extraction, association rule mining, unstructured data processing, and time-series forecasting.

Skill Extraction and Semantic Matching: The incorporation of Natural Language Processing (NLP) for skill identification has been extensively explored. Zhang et al. (2022) [6] have shown the viability of weak supervision paradigms coupled with transformer models like RoBERTa and JobBERT for semantic matching of unstructured job descriptions to standard taxonomies like ESCO. In the absence of non-explicit requirements, researchers like Gugnani and Misra (2020) [7]have successfully employed Doc2Vec for implicit skills extraction from job descriptions. However, comprehensive surveys by Senger et al. (2023) [8] and Khaouja et al. (2021) [9]reveal that although deep learning models are highly proficient in identifying existing skills, the temporal dynamics of future skills is still an area of concern.

Topic Modelling and Association Rule Mining: In order to understand the grouping of skills, Li et al. (2025) [10] utilized the BERTopic-TOPSIS model to cluster job description topics and determine the importance of skills based on factors such as salary. In the context of Sri Lanka, Kuruppu and Galappaththi (2020) [4] employed algorithms such as Apriori and FP-Growth to identify the frequent associations between job titles and technology from images of job vacancies. In addition, Chamithri and Asanka (2024) [5] proposed an NLP framework based on TF-IDF and Generative AI to identify Knowledge, Skills, and Abilities (KSAs) in Sri Lanka. However, the authors clearly state that their research was based on the time frame of June to November 2023 and thus emphasize the need for future research in the area of future skill demands.

Handling Unstructured Image Data (Dark Data): In the local market environment, job vacancies are often communicated through social media flyers. Dergachov et al. (2020) [11] have shown the accuracy of Tesseract OCR and OpenCV algorithms. In the Sri Lankan IT job market environment, Kuruppu and Galappaththi (2020) [4] have successfully applied the above algorithms to extract text from 500 job vacancy images for the purpose of trend analysis. Although the above approach has overcome the image-text challenge, it is only based on static trends.

Predictive Time Series Forecasting: Chen et al. (2024) [12] sought to bridge the predictive gap by presenting the Job SDF dataset, where several models, including ARIMA, LSTM, and Fourier-based models, were benchmarked for predicting monthly skill demands. The study demonstrates that statistical time series modeling, such as using ARIMA, is extremely effective for modeling momentum, seasonality, and structural changes in evolving skill sets. Wanyama (2025) [13] successfully implemented ARIMA and Holt-Winters for predicting emerging IT skill demands in Uganda, presenting a concrete methodology for similar emerging nations. Significantly, Ndungi and Blekanov (2025) [14] used a sliding window methodology to validate

that short-term forecasting indeed yields the lowest error rates, thereby scientifically proving that a 3-month forecasting window is indeed the most scientifically defensible.

1.2.2 The Spatial Dimension: Global vs. Local Markets

Existing literature on the development of IT economies mostly focuses only on the local job market environment. Studies of the Southeast Asian economy (Kuruppu & Galappaththi, 2020) [4] and the Sri Lankan economy (Chamithri & Asanka, 2024) [5] only scrape the job portals of the region. However, the IT industry is a very globalized sector. Emerging technologies tend to have spikes in the global remote job market (for example, the USA or European Union) before they are formally adopted by Sri Lankan corporate entities. Existing literature is completely devoid of a Global-Local Lead-Lag Analysis. The existing literature does not utilize the global remote job market as a predictive leading indicator of the local job market. It is a huge opportunity missed by the existing literature.

1.2.3 Review of Commercial Products and Services

The commercial space for skill intelligence seems to be dominated by platforms that provide retrospective analysis rather than accessible, predictive capabilities:

Global Platforms (LinkedIn, Indeed): LinkedIn's Economic Graph Reports (2024) [15] aggregate data to identify skills that have grown at a high rate in the past year. This methodology, however, is by necessity retrospective: a skill will only be included in such a report once that skill has already saturated the market, which is too late for an undergraduate looking to plan his/her upskilling strategy.

Local Job Portals (TopJobs.lk, XpressJobs): These platforms offer real-time visibility into current vacancies, but zero longitudinal analysis or prediction capabilities.

Enterprise Platforms (Lightcast): International enterprise platforms such as Lightcast (2024) [16] offer sophisticated labor market analytics but are geared towards institutional use at a prohibitively expensive cost for individual students.

This lack of accessible, predictive tooling requires undergraduates to speculate about their future learning requirements. This phenomenon has been strongly substantiated by our primary research survey, in which 89.4% of Sri Lankan IT undergraduates confirmed that they would derive immense value from a tool that would predict what skills would be in demand in the near future.

1.2.4 Synthesis of Identified Limitations and Research Gap

Synthesizing the reviewed literature and commercial offerings reveals three persistent limitations that this research specifically addresses:

Table 3:Synthesis of Identified Limitations and Research Gap

Research Domain	Existing Methods	Technical Gap / Limitation	Requirement for This Component
Skill Extraction	OCR-based text recognition (Tesseract/OpenCV); TF-IDF, Apriori	Focuses solely on extraction accuracy or static association rules; lacks semantic understanding of synonyms	Semantic Clustering: BERTopic to normalize synonyms and map complex IT jargon
Temporal Analysis	Static descriptive analytics; retrospective snapshots	No short-term momentum forecasting; students cannot distinguish rising vs. falling skills	Predictive Intelligence: ARIMA/Exponential Smoothing for 3-month market velocity forecasts
Spatial Analysis	Isolated local market studies; no cross-market comparison	Ignores predictive value of global trends as leading indicators for local adoption	Global-Local Lead-Lag: Cross-correlation and Granger causality to quantify time lags

1.2.5 Comparison of Existing Literature and Proposed System

Table 4 Comparison of Existing Literature and Proposed System

Author / System	Primary Approach / Technology	Analytics Focus	Data Ingestion Type	Key Limitations / Gaps Identified
Zhang et al. (2022) / Senger et al. (2023) [6]	NLP, Weak Supervision, RoBERTa	Descriptive (Static)	Structured Text	Excels at identifying current skills but lacks temporal forecasting for emerging market trends.
Li et al. (2025) / Gugnani & Misra (2020) [17]	BERTopic, TOPSIS, Doc2Vec	Descriptive (Static)	Structured Text	Provides excellent semantic clustering and implicit skill mapping but represents only a historical snapshot.
Kuruppu & Galappaththi (2020) [4]	OCR, Apriori, TF-IDF	Descriptive (Static)	Unstructured Image Data	Successfully processes image flyers but relies on basic association rules without predictive capability.
Chamithri & Asanka (2024) [5]	TF-IDF, Generative AI	Descriptive (Static)	Structured Text	Sri Lankan context, but exclusively focused on current demand (June-Nov 2023). Explicitly recommends predictive future work.

Chen et al. (2024) [12]	ARIMA, LSTM, Transformers	Predictive	Structured Text	Strong predictive modelling of job skills, but suffers from text-bias and is tailored to Chinese/global markets, ignoring Sri Lankan market lag.
Commercial Portals (LinkedIn, TopJobs) [15]	Keyword Matching, Search Indexing	Reactive / Historical	Structured Text	Rigid matching; ignores semantic variations, lacks predictive career modelling, and is blind to non-searchable image-based posts.
Proposed Solution	**OpenCV/OCR + BERTopic + ARIMA**	**Predictive (3-6 Months)**	**Hybrid (Image + Text)**	**Overcomes text-bias by digitising flyers, groups skills semantically, and provides proactive forecasting for students.**

1.2.6 Conclusion of Literature Review

However, the overarching limitation of all existing research and commercial systems is the Intelligence Lag: they are excellent at descriptive intelligence, i.e., finding what is currently in demand, but they do not capture the temporal nature of skills or use global trends as predictive features. Also, all existing systems are plagued by Text-Bias: they only use structured text data and completely ignore image-based dark data (flyers) that is so common in local recruitment.

In overcoming these semantic, temporal, and spatial limitations, this research puts itself at the cutting edge of predictive career analytics, offering a unique and highly practical system directly relevant to the needs of Sri Lankan IT undergrads.

1.3 Research Gap

A critical analysis of the existing literature and commercial offerings indicates that although the need for IT skill analytics has been recognized, the methodologies currently adopted do not offer effective proactive, localized, and context-aware advice to undergraduates. The current methods are not effective due to three major reasons, which are the core research gap addressed in this research.

1.3.1. The Semantic Extraction Gap

Currently, in local research, syntactic-based text mining techniques such as Term Frequency-Inverse Document Frequency (TF-IDF) and the Apriori algorithm (Kuruppu & Galappaththi,

2020) [4] are used. These techniques, though traditional and commonly used in text mining, are insufficient as they lack semantic analysis. These techniques only allow for exact word matching and cannot interpret that words such as NodeJS, Node.js, and Express relate to the same category of IT skills. Moreover, in cases where image-based job advertisements on various mediums such as flyers posted on local social media platforms are used, Optical Character Recognition (OCR) noise is high, and syntactic-based text mining techniques cannot handle this type of data well. Thus, a gap is identified in using deep learning-based semantic clustering models for accurate normalization of complex IT jargon in Sri Lanka (Senger et al., 2023 [18]).

1.3.2. The Temporal Forecasting Gap

Secondly, existing career analytics tools and research works are entirely retrospective in nature (Chamithri & Asanka, 2024) [5]. These tools and works only carry out static descriptive analytics, i.e., they only show us what skills were in demand in the past and not what skills will be in demand in the future. This is highly insufficient for undergraduates who have to prepare for future recruitment drives. Although recent global research works have successfully used time series forecasting (e.g., ARIMA) for job markets in other developing nations (Wanyama, 2025) [13], short-term momentum forecasting has not been incorporated into a localized tool for Sri Lankan students. This makes students oblivious of market velocity and hence makes them rely on old data when making important learning decisions.

1.3.3. The Spatial Analysis Gap (Global vs. Local)

Thirdly, the existing literature portrays the Sri Lankan IT job market as an independent environment. In the existing literature, researchers only study the local job portals. However, the IT sector is a highly globalized environment. It is an unfortunate oversight because the literature shows that new technologies often have a spike in the global remote job market before they are formally adopted by local corporate job markets. In fact, this is well documented in the technology diffusion literature (Rogers, 1962/2010 [19], Bloom et al., 2021 [20]). At present, no literature uses mathematical cross-correlation of data from the foreign job market to predict these changes. Sri Lankan students are unaware of the potential of global job market trends.

1.3.4 Justification of Novelty and Creativity

In order to address these gaps, which were clearly identified, a novel Semantic Forecasting Engine has been conceived as part of this research component. The creativity of this solution resides in the tri-fold application of advanced Data Science methodologies, which were not previously combined for this target audience.

Semantic Over Syntactic: It uses the BERTopic approach coupled with a custom noise reduction pipeline for OCR data to enable the extraction of semantic skills. This effectively circumvents the shortcomings of the conventional approach of matching keywords.

Proactive Over Retrospective: It changes the paradigm from descriptive to predictive analytics by using Exponential Smoothing and ARIMA models to forecast 3-month tactical market

velocity. The short-term forecast of 3 months is mathematically validated by the literature (Ndungi & Blekanov, 2025) [14], which shows that beyond 3 months of data, significant errors are introduced when trying to forecast the labor market.

Global-Local Lead-Lag Analysis: This feature creates a highly innovative spatial analysis tool. By comparing global remote data sets with local Sri Lankan data sets through Cross-Correlation Function (CCF) and Granger Causality Tests, quantifiable lag times are determined. This serves as an early warning radar for undergraduates seeking a strategic first-mover advantage for new global trends prior to market saturation in the local market.

1.3.5 Evidence-Based Argument for the Research Gap

The necessity and creativity of this approach are not only theoretical but are strongly validated by empirical evidence. Indeed, as deduced from Question 3 of our primary survey of 320 IT undergraduates, an overwhelming 89.4% of IT undergraduates face the problem of mismatch between the curriculum of the university they attend and the job requirements.

Moreover, since existing tools are only useful for analyzing historical data, the vast majority of the students reported extreme difficulties in accurately predicting what technologies they should be trained in to secure jobs by the time of graduation. By filling the semantic, temporal, and spatial gaps of existing research, we offer a highly practical solution to a proven educational crisis.

By filling the semantic, temporal, and spatial gaps in current research, this component provides a highly practical, data-driven solution that directly addresses a proven national educational crisis.

2. OBJECTIVE

2.1 Main Objective

To design, develop, and mathematically evaluate an AI-driven **Semantic Skill Forecasting Engine** that empowers Sri Lankan IT undergraduates to bridge the industry-academia skill gap by providing highly accurate 3-month skill momentum predictions through the integration of deep-learning-based semantic skill extraction and Global-Local Lead-Lag Analysis.

2.2 Specific Objectives

In order to successfully realize the primary objective of creating the Semantic Skill Forecasting Engine, the overall process of researching the proposed topic has been broken down into five highly measurable specific objectives. These specific objectives are intended to directly resolve the semantic, temporal, and spatial gaps in the literature while resolving the issues of predictability raised by the undergraduate students surveyed for this report.

Table 5 SMART Breakdown of Specific Objectives

Objective	SMART Breakdown
1. To engineer a robust Data Acquisition and Preprocessing Pipeline that automatically extracts unstructured text from both standard web-based job portals and image-based promotional flyers with a target Character Error Rate (CER) below 10%.	**Specific:** Builds OCR pipeline **Measurable:** CER < 10% **Achievable:** Tesseract/OpenCV proven **Relevant:** Addresses dark data gap **Time-bound:** Completed by PP1
2. To implement and fine-tune a Semantic Topic Modeling framework (BERTopic) that achieves at least 85% precision and recall in clustering synonymous IT skills (e.g., grouping NodeJS, Node.js, and Express) when validated against a manually annotated dataset of 500 job postings.	**Specific:** BERTopic implementation **Measurable:** Precision/Recall ≥ 85% **Achievable:** BERTopic proven in literature **Relevant:** Addresses semantic gap **Time-bound:** Completed by PP1
3. To formulate and execute a Global-Local Lead-Lag Analysis that identifies statistically significant time delays ($p < 0.05$) between skill demand spikes in global remote markets and subsequent adoption in Sri Lanka, using Cross-Correlation Function (CCF) and Granger Causality tests on 12 months of paired time-series data.	**Specific:** CCF + Granger tests **Measurable:** $p < 0.05$ significance **Achievable:** 40+ weekly data points **Relevant:** Addresses spatial gap **Time-bound:** Completed by PP2
4. To develop and back-test Short-Term Time-Series Forecasting Models (ARIMA and Exponential Smoothing) that predict 3-month skill demand velocity with Mean Absolute Percentage Error (MAPE) below 20%, validated using sliding-window cross-validation on 12 months of historical data.	**Specific:** ARIMA + ETS models **Measurable:** MAPE < 20% **Achievable:** Matches literature benchmarks **Relevant:** Addresses temporal gap **Time-bound:** Completed by PP2
5. To integrate forecasting outputs and lead-lag insights into an Interactive Analytical Dashboard that achieves a System Usability Scale (SUS) score of at least 70 when tested with 15 target users (IT undergraduates).	**Specific:** Dashboard development **Measurable:** SUS score ≥ 70 **Achievable:** Standard usability target **Relevant:** Ensures practical usability **Time-bound:** Completed by final submission

3.METHODOLOGY

3.1 Research Approach

This study follows Design Science Research (DSR) as the primary research methodology, guided by the process model developed by Peffers et al. (2007). DSR is particularly relevant for this component of the research as it is not only an observation of a phenomenon but also the development and evaluation of a novel technological artifact, namely a Temporal Skill Demand Forecasting Engine, aimed at combating intelligence lag in existing career guidance systems.

The DSR methodology is a robust framework for the development and evaluation of this predictive engine through its six phases:

Table 6 Application in This Research

Phase	Application in This Research
Problem Identification	Demonstrated through survey data (89.4% skill mismatch) and literature gaps
Objective Definition	SMART objectives defined in Section 2.2
Design & Development	Building OCR pipeline, BERTopic clustering, ARIMA models, lead-lag analysis
Demonstration	Testing the engine with real job posting data
Evaluation	Quantitative validation using CER, Precision/Recall, MAPE, *p*-values, SUS
Communication	Proposal, final report, and presentations

3.2 Technical Methods and Algorithms

The core technical methodology integrates semantic clustering with econometric time-series modeling to move from descriptive snapshots to predictive foresight.

Table 7 Technical Validation Metrics

Method	Purpose	Implementation Details
OCR Text Extraction	Digitize image-based job flyers (dark data)	Tesseract OCR + OpenCV for image preprocessing (thresholding, noise reduction)
Text Cleaning	Normalize extracted text	Regular expressions, stop-word removal, stemming
Semantic Clustering	Group synonymous skills (e.g., NodeJS = Node.js)	BERTopic with sentence-transformers for embeddings; class-based TF-IDF for topic representation
Time-Series Construction	Convert skill mentions to temporal format	Weekly aggregation using Pandas; create matrices of skill frequency over time
Lead-Lag Analysis	Quantify time delays between global and local trends	Cross-Correlation Function (CCF) to identify optimal lag; Granger Causality tests ($p < 0.05$) to establish predictive relationship

Forecasting	Predict 3-month skill demand	ARIMA (AutoRegressive Integrated Moving Average) and Exponential Smoothing (Holt-Winters)
Model Selection	Identify optimal parameters	Augmented Dickey-Fuller (ADF) test for stationarity; Akaike Information Criterion (AIC) for ARIMA(p,d,q) selection
Visualization	Present insights to users	Interactive dashboard using Plotly/Dash; time-series plots, lag visualizations, confidence intervals

3.3 Tools and Platforms with Justification

Table 8 Tools and Platforms with Justification

Tool / Platform	Purpose	Justification	Alternatives Considered
Python 3.9+	Primary programming language	Rich ecosystem for NLP, time-series, and statistical analysis	R (weaker integration with OCR and web frameworks)
Tesseract OCR	Text extraction from images	Open-source, proven accuracy, Python bindings (pytesseract)	Google Vision API (paid, not feasible for research)
OpenCV	Image preprocessing	Industry standard for noise reduction and thresholding	PIL/Pillow (less powerful for complex preprocessing)
BERTopic	Semantic skill clustering	State-of-the-art for topic modeling; handles synonyms via embeddings	LDA (bag-of-words, no semantic understanding)
Pandas / NumPy	Data manipulation and aggregation	Standard for time-series data processing	Excel (not scalable for 1000+ job postings)
Statsmodels	ARIMA, CCF, Granger causality	Comprehensive statistical library with peer-reviewed implementations	scikit-learn (limited time-series support)
Matplotlib / Plotly	Visualization	Interactive dashboards for user insights	Tableau (costly, less customizable)
Jupyter Notebooks	Development and experimentation	Iterative testing and visualization	VS Code (less interactive for exploration)

3.4 Data Collection and Ethical Considerations

Table 9 Data Collection

Aspect	Details
Data Sources (Local)	TopJobs.lk, Xpress.jobs, lkman.lk – leading Sri Lankan job portals

Data Sources (Global)	LinkedIn Global, Indeed US/UK, Remote OK, We Work Remotely
Time Period	12 months of historical data
Collection Method	Ethical web scraping using BeautifulSoup/Scrapy with rate limiting (1 request/second) and respect for robots.txt
Image Data	Job flyers from social media (public pages) – processed via OCR
Sample Size Target	1000+ local job postings, 5000+ global postings
Data Storage	CSV files (anonymized, no PII) stored on secure university drive

Ethical Considerations

Table 10 Ethical Considerations

Principle	Application
No PII Collection	Only job descriptions, titles, and required skills – no applicant data
Public Domain Only	All data sourced from publicly accessible websites and social media pages
Respect for robots.txt	Automated scrapers will follow site policies and use polite delays
Informed Consent (Survey)	Survey respondents were anonymized; participation was voluntary
Data Security	All data stored on password-protected university systems
ICTA Guidelines	Adherence to Sri Lanka's data privacy norms and ICTA recommendations

3.5 Validation Strategy and Performance Metrics

The predictive accuracy of the artifact is evaluated using a multi-layered validation strategy aligned with the SMART objectives.

3.5.1 Technical Validation Metrics

Table 11 Technical Validation Metrics

Objective	Metric	Target	Validation Method
OCR Pipeline	Character Error Rate (CER)	< 10%	Compare OCR output against manually transcribed sample of 50 job flyers

BERTopic Clustering	Precision, Recall, F1-Score	≥ 85%	Manually annotated dataset of 500 skill mentions; compare cluster assignments
Lead-Lag Analysis	p-value (Granger Causality)	$p < 0.05$	Statistical significance test on paired global-local time series
Forecasting	Mean Absolute Percentage Error (MAPE)	< 20%	Sliding-window cross-validation on 12 months of historical data
Forecasting	Root Mean Square Error (RMSE)	Minimize	Compare predicted vs. actual values in test set
Dashboard	System Usability Scale (SUS)	≥ 70	User testing with 15 IT undergraduates

4. HIGH-LEVEL SYSTEM ARCHITECTURE

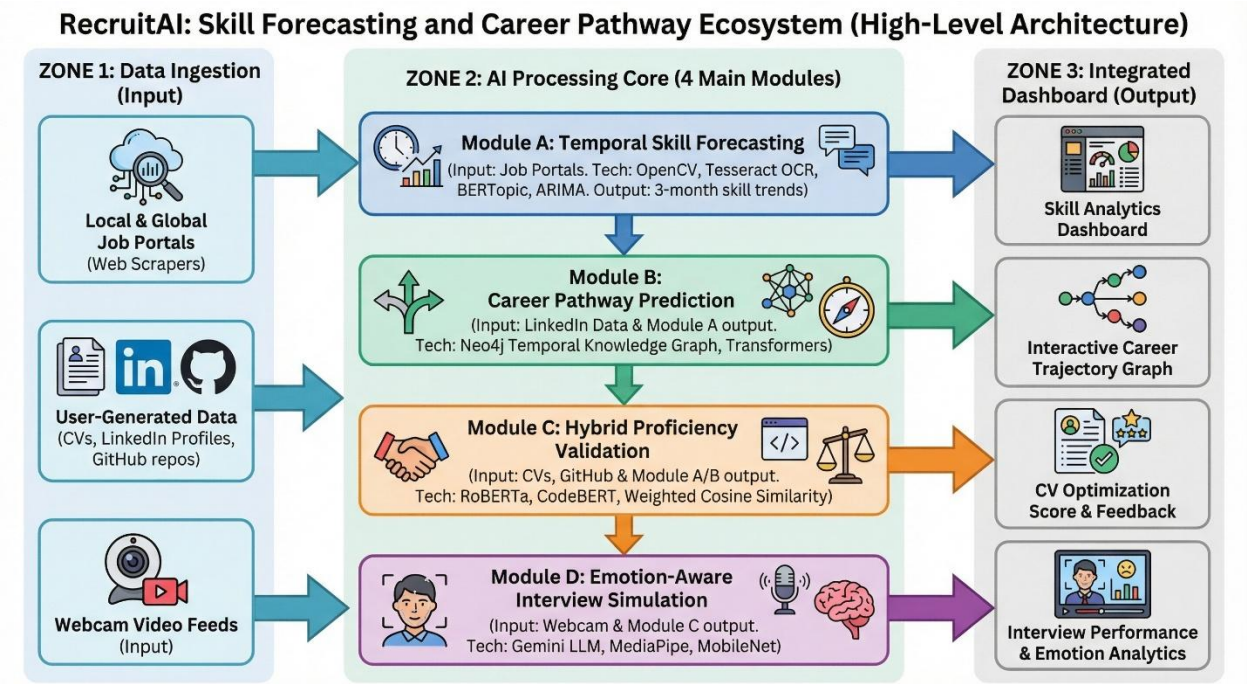


Figure 0-4 High-Level System Architecture

4.1 Overview of Proposed Solution

The proposed AI-Powered Career Intelligence Ecosystem is a unified platform that combines four primary components to address the skill gap for Sri Lankan IT undergraduates. It leverages

raw job market data (structured text, image-based flyers, and global remote work listings) and transforms it into meaningful insights through:

1. Skill Forecasting Engin– predicts future skills and measures global-local adoption lags.
2. Career Pathway Predictor – predicts career paths using graph databases and sequence prediction.
3. CV Optimizer – performs semantic matching of resumes and job requirements and provides optimization suggestions.
4. Interview Simulator – tracks user attention in real-time using LLM and computer vision techniques.

All components utilize a unified data pipeline and dashboard, facilitating seamless user interaction.

4.2 System Diagram

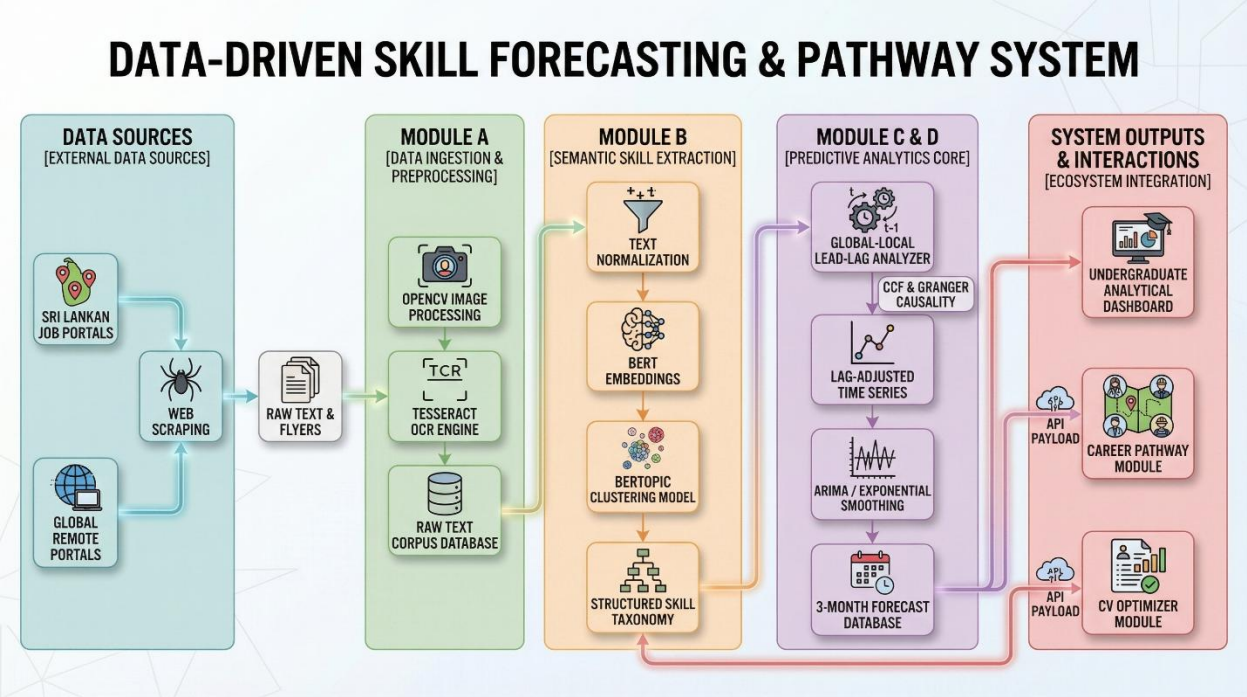


Figure 0-5 Data-Driven Skill Forecasting & Pathway System

4.3 Highlight of Individual Component and Its Parts

The **Skill Forecasting Engine**, consists of the following sub-modules:

1. **Data Acquisition Layer** – scrapes job postings from local (TopJobs, Xpress) and global (LinkedIn, Remote OK) sources and captures image-based flyers via a custom crawler.
2. **OCR Preprocessor** – uses OpenCV and Tesseract to digitise image flyers and clean extracted text.
3. **Semantic Clustering (BERTopic)** – groups synonymous skills (e.g., NodeJS and Express) into meaningful clusters.
4. **Time-Series Constructor** – aggregates skill mentions by week, creating temporal profiles for each skill cluster.
5. **Lead-Lag Analyzer** – applies Cross-Correlation Function (CCF) and Granger causality tests to quantify time delays between global and local trends.
6. **Forecaster** – trains ARIMA and Exponential Smoothing models to generate 3-month demand predictions.
7. **Output Visualiser** – delivers interactive charts and alerts to the front-end dashboard.

4.4 Component Interactions

All components push their results to a shared analytics database, from which the front-end dashboard retrieves personalized recommendations.

Table 12 Component Interactions

Component	Interacts With	Data Flow
Skill Forecasting Engine	All other modules	Supplies skill demand forecasts and emerging skill lists to the Career Pathway Predictor and CV Optimizer.
Career Pathway Predictor	Skill Forecasting Engine	Uses forecasted skills to refine recommended career paths.
CV Optimizer	Skill Forecasting Engine	Compares user skills against forecasted demand to highlight gaps.
Interview Simulator	None directly; standalone	Uses its own LLM and vision pipeline, but can be enhanced with forecasted skill themes.

4.5 Data Flow Explanation

1. **Ingestion** – Raw job postings are ingested every day in the form of HTML, JSON, and images.
2. **Processing** – The skill forecasting engine processes the new data, applies OCR and BERTopic, and updates time series databases.

3. **Forecast Generation** – Every week, the engine uses ARIMA models for each skill cluster, stores the results, and does lead-lag analysis.
4. **Consumption** – The dashboard queries the previously generated results for:
5. Current top skills
6. 3-month forecasts with confidence intervals
7. Early warnings for global skills expected to arrive locally
8. Feedback – User feedback is collected based on their interactions with the dashboard, for example, which alerts they click on.

4.6 Scalability and Security Considerations

Scalability:

Horizontal Scaling: OCR and NLP processes are stateless and can be scaled horizontally by distributing them across multiple containers (Docker/Kubernetes).

Incremental Updates: Rather than re-running all data weekly, new posts are incrementally added, and forecasts are recalculated only for affected skill clusters.

Caching: Frequently accessed results (e.g., results from the previous month's forecasts) can be cached to avoid overloading the database.

Security:

Data Privacy: No personally identifiable information (PII) is stored. All scraped data is anonymized.

Access Control: Role-based access (student, admin) will be implemented via API tokens.

Secure API: HTTPS and API keys are used for communication between microservices.

Rate Limiting: Scrapers are designed to respect the robots.txt file of the websites and use polite delays to avoid overwhelming the server.

Audit Logs: All data access and model runs are logged for compliance and debugging.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The primary stakeholders are identified as IT undergrads in Sri Lanka. This has been confirmed through our primary survey of 320 students

- 89.4% have identified a mismatch between university curriculum and job requirements.
- 56% are currently in search of a job.
- 87.5% find it difficult to predict which skills are likely to be in demand in 6 months' time.
- 100% are willing to adjust their plans if a reliable 6-month forecast were available.

The secondary stakeholders are:

Universities & Curriculum Planners – need data to ensure courses are aligned to meet new industry requirements.

Career Guidance Units – need real-time, personalized tools to assist students in making choices.

IT Employers – need graduates who possess relevant skills in a rapidly changing industry.

ICTA & Policymakers – need data to support workforce development initiatives.

5.2 Functional Requirements

The Skill Forecasting Engine system must deliver the following capabilities:

- Automatically scrape and ingest at least three job portals from Sri Lanka (TopJobs.lk, Xpress.jobs, Ikman.lk) and at least two global remote job portals (LinkedIn, Remote OK).
- Extract text from image-based job flyers via Optical Character Recognition (OCR) with a target Character Error Rate (CER) of less than 10%.
- Achieve semantic clustering of technical skills via BERTopic with at least 85% precision and recall on a manually annotated dataset.
- Derive weekly time series for each skill cluster for both local and global markets.
- Calculate lead-lag relationships between global and local skill trends via cross-correlation and Granger causality ($p < 0.05$).
- Make 3-month demand forecasts for each skill cluster via ARIMA and exponential smoothing with a target Mean Absolute Percentage Error (MAPE) of less than 20%.

- Develop an interactive dashboard displaying current skill trends, 3-month forecasts, and early warning signals for globally emerging skills.

5.3 Non-functional Requirements

These requirements directly address the needs identified in the survey and ensure the system is both technically robust and practically useful for its intended users.

Table 13 Non-functional Requirements

Category	Requirement	Target
Usability	The dashboard must achieve a System Usability Scale (SUS) score of at least 70 when tested with 15 target undergraduates.	Ensures the interface is intuitive and actionable.
Performance	Forecast generation for a single skill cluster should complete within 30 seconds on standard hardware.	Supports weekly retraining without delaying user experience.
Performance	Dashboard page load time must be ≤ 3 seconds on a typical broadband connection.	Maintains user engagement.
Security	No personally identifiable information (PII) shall be stored. All scraped data must be anonymised.	Complies with ethical guidelines and university policy.
Security	API endpoints must be protected by authentication tokens; dashboard access must be role-based (student, admin).	Prevents unauthorised data access.
Security	All inter-service communication must use HTTPS	Protects data in transit.
Scalability	The system must support incremental updates (new job postings added without full recomputation).	Enables weekly forecasts without excessive processing overhead.
Reliability	The forecasting pipeline must maintain ≥ 95% uptime during the evaluation phase.	Ensures continuous availability for user testing.
Maintainability	All modules must be containerised (Docker) with documented APIs.	Facilitates future enhancements and integration.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

Target Market and Customer Persona,

Primary: Final-year IT undergraduates and fresh graduates in Sri Lanka preparing for job applications and technical interviews.

Secondary: Academic institutions (universities, career guidance units) seeking data-driven tools to align curricula with real-time market shifts.

Value Proposition

A locally adapted AI environment offering:

- Skill forecasting (3-month demand forecasts) and global-local trend analysis.
- Semantic CV-job matching, moving beyond keyword searching.
- Interview simulations, including real-time attention and emotion feedback.
- Visualization of career pathways based on real Sri Lankan professional data.

This minimizes graduates' bench time (the need for retraining after recruitment) and helps institutions and students stay at the cutting edge of rapidly changing IT requirements.

Revenue Model

Table 14 Revenue Models

Model	Description
Premium	Basic skill analytics and career graph free; advanced features (full forecast, CV optimizer, interview simulator) via monthly subscription.
Subscription	Students pay monthly or annually for full platform access.
School Licensing	Annual B2B license for universities and career guidance units; includes institutional dashboard for curriculum planning.
Educational Partnerships	Collaborations with HR firms, and industry partners to integrate the platform into placement and employability programs.

Cost Estimation

Table 15 Estimated Costs

Item	Estimated Cost (LKR)
Cloud infrastructure (AWS/DO)	150,000
Database & API hosting	100,000
Developer tools & licenses	50,000
Marketing & user acquisition	200,000
Total	500,000

Pricing Strategy

- Student subscription: LKR 1000/month or LKR 10,000/year.

- University license: LKR 500,000/year (unlimited student access + analytics dashboard).
- Freemium conversion: Target 5% of free users converting to paid after 30 days.

Competitive Advantage

Localization: Using Sri Lankan job ads and career histories instead of general global data.

Predictive, not Descriptive: Using ARIMA-based 3-month forecasts to give students an edge over the competition.

Unified Ecosystem: Integrating forecasting, CV optimization, and behavioral interview simulation into one system.

Ethical Data Collection: Transparent, privacy-respecting, and complying with website terms.

Intellectual Property Considerations

- Trade secrets: Proprietary algorithms for weighted matching of skills, BERTopic-based clustering of skills, and Neo4j-based models for career pathways.
- Original IP: NLP models fine-tuned for Sri Lankan IT vocabulary and integrated feedback loops.
- Data ownership: All data is publicly sourced, no personal data is collected from users.
- IP Protection: Plans for future utility patent for global-local lead-lag methodology and software copyright.

7. BUDGET AND JUSTIFICATION

The following budget outlines the estimated costs for implementing and testing the proposed AI-driven career intelligence ecosystem. All figures are in Sri Lankan Rupees (LKR). The budget is based on a 6-month development and evaluation period with four team members.

Table 16 Budget Summary

Category	Justification	Amount (LKR)
Hardware Costs	Webcams for real-time attention tracking in the interview simulation module (using MediaPipe / FER-2013) and an external SSD for storing scraped job posting datasets.	40,000
Software Licenses	All tools are free and open-source: Python, React, Node.js, spaCy, Tesseract OCR, MediaPipe, Neo4j Community Edition, Hugging Face Transformers, GitHub, VS Code.	0
Cloud Services	Google Cloud Run (for OCR/NLP pipeline), Gemini API (interview simulation), MongoDB Atlas, Firebase (dashboard authentication), Vercel / Render (deployment).	65,000
Human Effort Cost	Market-rate contribution of 4 members × 120 hours, LKR 250/hour each, for academic reporting purposes. No cash salaries are disbursed.	120,000
Miscellaneous	Internet costs for ethical web scraping (TopJobs.lk, Xpress.jobs, lkman.lk), API calls, and cloud access throughout the project duration.	15,000

Total Estimated Budget

LKR 240,000

8. WORK BREAKDOWN STRUCTURE (WBS)

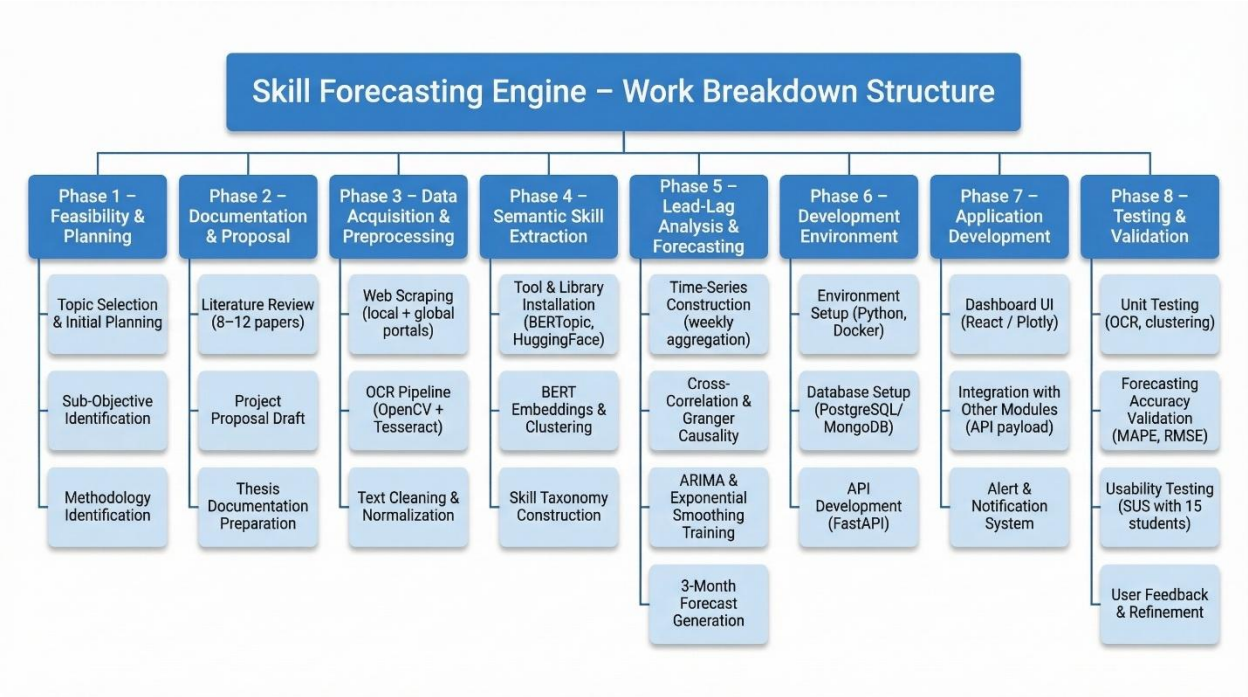


Figure 0-18. WORK BREAKDOWN STRUCTURE (WBS) DIAGRAM

9. GANTT CHART

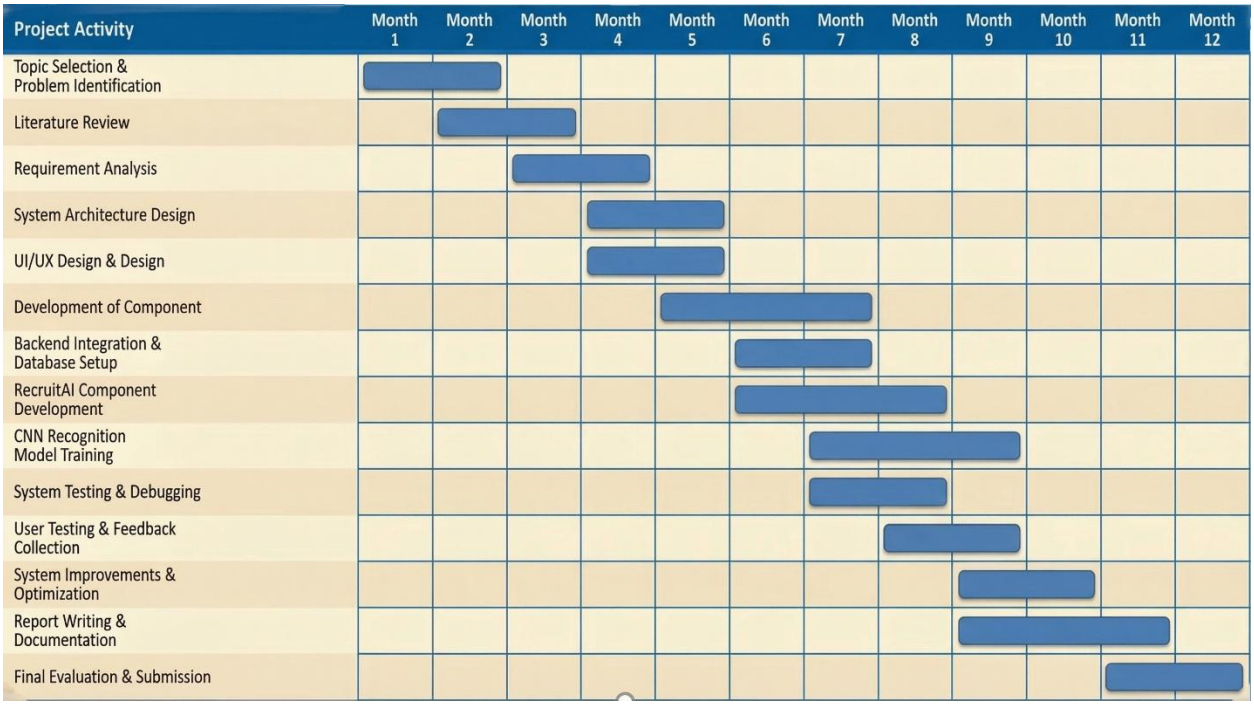


Figure 0-1 GANTT CHART

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